

Linear regression

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
y = np.array([2, 4, 5, 4, 5])
model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)
# Plot
plt.scatter(X, y)
plt.plot(X, y_pred)
plt.title("Linear Regression")
plt.show()
```

Logistic regression

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
X = np.array([1,2,3,4,5,6,7,8]).reshape(-1,1)
y = np.array([0,0,0,0,1,1,1,1])
model = LogisticRegression()
model.fit(X, y)
X_test = np.linspace(1, 8, 100).reshape(-1,1)
y_prob = model.predict_proba(X_test)[:,-1]
plt.scatter(X, y)
```

```
plt.plot(X_test, y_prob)
plt.title("Logistic Regression")
plt.show()
```

Decision Tree

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt

data = load_iris()
X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
print("Accuracy:", model.score(X_test, y_test))

plt.figure(figsize=(8,6))
plot_tree(model, filled=True)
plt.show()
```

Naive Bayes (Gaussian)

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

data = load_iris()
X = data.data
```

```
y = data.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = GaussianNB()
model.fit(X_train, y_train)
print("Accuracy:", model.score(X_test, y_test))
```

PCA (reduce to 2D + plot)

```
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

data = load_iris()
X = data.data
y = data.target

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

plt.scatter(X_pca[:,0], X_pca[:,1], c=y)
plt.title("PCA - 2D")
plt.show()
```

Random Forest (accuracy)

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

data = load_iris()
X = data.data
y = data.target
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = RandomForestClassifier()
model.fit(X_train, y_train)
print("Accuracy:", model.score(X_test, y_test))
```